

Differentiation & Linear Algebra

Single-column formula reference

I Differentiation — Core Rules

Definition of the derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Sum rule

$$(f \pm g)' = f' \pm g'$$

Product rule

$$(fg)' = f'g + fg'$$

Quotient rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Chain rule

$$(f \circ g)'(x) = f'(g(x)) g'(x)$$

Inverse function rule

$$\frac{d}{dx} [f^{-1}(x)] = \frac{1}{f'(f^{-1}(x))}$$

Power rule

$$\frac{d}{dx} [x^n] = nx^{n-1}$$

Exponential functions

$$\frac{d}{dx} [e^x] = e^x, \quad \frac{d}{dx} [a^x] = a^x \ln a$$

Logarithmic functions

$$\frac{d}{dx} [\ln x] = \frac{1}{x}, \quad \frac{d}{dx} [\log_a x] = \frac{1}{x \ln a}$$

Trigonometric derivatives

$$\frac{d}{dx}[\sin x] = \cos x, \quad \frac{d}{dx}[\cos x] = -\sin x, \quad \frac{d}{dx}[\tan x] = \sec^2 x$$

Inverse trigonometric derivatives

$$\frac{d}{dx}[\arcsin x] = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}[\arctan x] = \frac{1}{1+x^2}$$

Hyperbolic derivatives

$$\frac{d}{dx}[\sinh x] = \cosh x, \quad \frac{d}{dx}[\cosh x] = \sinh x$$

II Differentiation — Advanced Topics

Leibniz rule (higher-order product derivative)

$$(fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}$$

Taylor series expansion

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

Faà di Bruno's formula

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{k=1}^n f^{(k)}(g(x)) B_{n,k}(g', g'', \dots, g^{(n-k+1)})$$

Implicit differentiation

$$\frac{dy}{dx} = -\frac{\partial F / \partial x}{\partial F / \partial y} \quad \text{for } F(x, y) = 0$$

Parametric differentiation

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Second derivative, parametric form

$$\frac{d^2 y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{dx/dt}$$

Gradient

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)$$

Total derivative

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i$$

Multivariable chain rule

$$\frac{\partial f}{\partial t} = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial t}$$

Jacobian matrix

$$J = \left[\frac{\partial f_i}{\partial x_j} \right]_{i,j}$$

Hessian matrix

$$H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

Mean Value Theorem

$$f'(c) = \frac{f(b) - f(a)}{b - a}, \quad c \in (a, b)$$

L'Hôpital's Rule

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Newton's method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

III Linear Algebra — Matrices & Determinants

Transpose of a product

$$(AB)^T = B^T A^T$$

Inverse of a product

$$(AB)^{-1} = B^{-1} A^{-1}$$

Inverse via adjugate

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

2×2 determinant

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

3×3 determinant (Sarrus' rule)

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

Cofactor (Laplace) expansion

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} M_{ij}$$

Determinant of a product

$$\det(AB) = \det(A) \det(B)$$

Determinant of an inverse

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

Trace of a matrix

$$\operatorname{tr}(A) = \sum_{i=1}^n a_{ii} = \sum_{i=1}^n \lambda_i$$

Cyclic property of trace

$$\operatorname{tr}(AB) = \operatorname{tr}(BA)$$

Eigenvalue equation

$$A\mathbf{v} = \lambda\mathbf{v} \iff \det(A - \lambda I) = 0$$

Determinant as product of eigenvalues

$$\det(A) = \prod_{i=1}^n \lambda_i$$

Diagonalization

$$A = PDP^{-1}, \quad D = \text{diag}(\lambda_1, \dots, \lambda_n)$$

Cayley–Hamilton theorem

$$p(\lambda) = \det(A - \lambda I) \implies p(A) = 0$$

IV Linear Algebra — Vector Spaces & Decompositions

Dot product

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

Vector norm

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}$$

Cauchy–Schwarz inequality

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

Cross product (in $\mathbb{R}^3$)

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Vector projection

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}$$

Rank–nullity theorem

$$\text{rank}(A) + \text{nullity}(A) = n$$

Gram–Schmidt orthogonalization

$$\mathbf{e}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \frac{\mathbf{v}_k \cdot \mathbf{e}_j}{\mathbf{e}_j \cdot \mathbf{e}_j} \mathbf{e}_j$$

LU decomposition

$$A = LU$$

QR decomposition

$$A = QR, \quad Q^T Q = I$$

Singular Value Decomposition

$$A = U \Sigma V^T$$

Spectral theorem (symmetric matrices)

$$A = Q\Lambda Q^T, \quad Q^T Q = I$$

Cramer's rule

$$x_i = \frac{\det(A_i)}{\det(A)}$$

Least squares solution

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$